

Vaibhav Randale

Full Stack Developer | MERN Stack | Scalable Backend Systems | Real-Time Applications | System Design | Node.js | Express.js
MongoDB | React.js | JavaScript | AWS | Redis | Queues | Workers | Rate Limiting | Middleware | Socket.IO | REST APIs
JWT Auth | BullMQ | Cron Jobs | MQTT | IoT | AI Agents | Subscriptions | Multi-Tenant | SaaS
vaibhavrandale800@gmail.com | +91 8007923255 | [LinkedIn](#) | [GitHub](#)

Summary

Full Stack Developer with 3 years and 6 months of experience building enterprise-grade MERN stack applications. Experienced in designing scalable backend systems with Redis caching, message queues, background workers, rate limiting, and custom middleware pipelines, alongside React.js frontend development for admin dashboards and client-facing portals. Expertise in real-time event-driven architecture, subscription-based workflows, RESTful API design, JWT authentication, RBAC, and ERP integrations serving 10+ enterprise clients with high-frequency data processing and AI-driven backend solutions.

Tools & Technologies

Languages: JavaScript (ES6+), SQL

Frontend: React.js, HTML5, CSS3

Backend: Node.js, Express.js, REST APIs, JWT Auth, RBAC, Middleware, Rate Limiting, Cron Jobs

Databases: MongoDB, MySQL, Redis (Caching)

Queues & Workers: BullMQ, Message Queues, Background Workers, Subscriptions

Real-Time & Protocols: Socket.IO, MQTT, LoRaWAN

Cloud & Deployment: AWS (EC2), Vercel, Render, Cloudinary

Tools: Claude Code, Cursor, Git, Postman, Httpie, VS Code

Integrations: WhatsApp Business API, OpenAI, ChirpStack, Cloudinary, NodeMailer

Concepts: System Design, Distributed Systems, Event-Driven Architecture, ERP Integration, AI Agents

Experience

Full Stack Developer · Taypro Pvt. Ltd., Pune, MH March 2023 - Present

Nectyr - B2B SaaS · Subscription-based · Multi-tenant · RBAC · 10+ Enterprise Clients

- Architected Nectyr, a production B2B SaaS platform – multi-tenant, subscription-based, serving 10+ enterprise clients with zero downtime.
- Built responsive React.js admin dashboards and client-facing modules consuming real-time Socket.IO data streams across 15+ feature modules.
- Engineered Redis caching and MongoDB query optimization, cutting API response time by 40% while sustaining throughput under high concurrent load.
- Built a distributed async job system using BullMQ workers, message queues, and subscription event handling – eliminating 10+ hrs/week of manual operations.
- Designed a real-time event pipeline (Socket.IO, message queues) delivering sub-second dashboard updates across high-frequency data streams.
- Engineered a secure middleware stack – rate limiting, JWT auth, granular RBAC – across 15+ modules.
- Architected a client-level subscription engine with plan-based feature gating, real-time validity checks, automated expiry enforcement, billing workflows, and WhatsApp-triggered expiry notifications across tiered SaaS plans.
- Integrated WhatsApp Business API for real-time notifications – robot failure alerts, subscription expiry warnings, and activity updates.
- Built an LLM-powered action layer using OpenAI's tool-calling API, routing natural language input to system actions via dynamic functions.
- Designed an ERP-integrated finance workflow with multi-level approval chains and automated fund disbursement sync, replacing a fully manual process.
- Developed a LoRaWAN ingestion pipeline (ChirpStack v4) with 99.5% uplink reliability and sub-second bidirectional command dispatch.

Personal Projects

- **Banking Application** - Full-stack platform with secure transactions, JWT auth, OTP verification, KYC, account management, and RBAC. (Stack: MERN, Redis, JWT)
- **E-commerce Platform** - Scalable store with PayPal integration, order lifecycle, dynamic search, and admin dashboard. (Stack: MERN, Cloudinary, NodeMailer)

Education

BE in Electronics & Telecommunication · PCCOE, Pune · CGPA: 9.37

Aug 2019 – June 2022

Diploma in Electronics · PLGP, Latur · 88.06%

July 2016 – May 2019